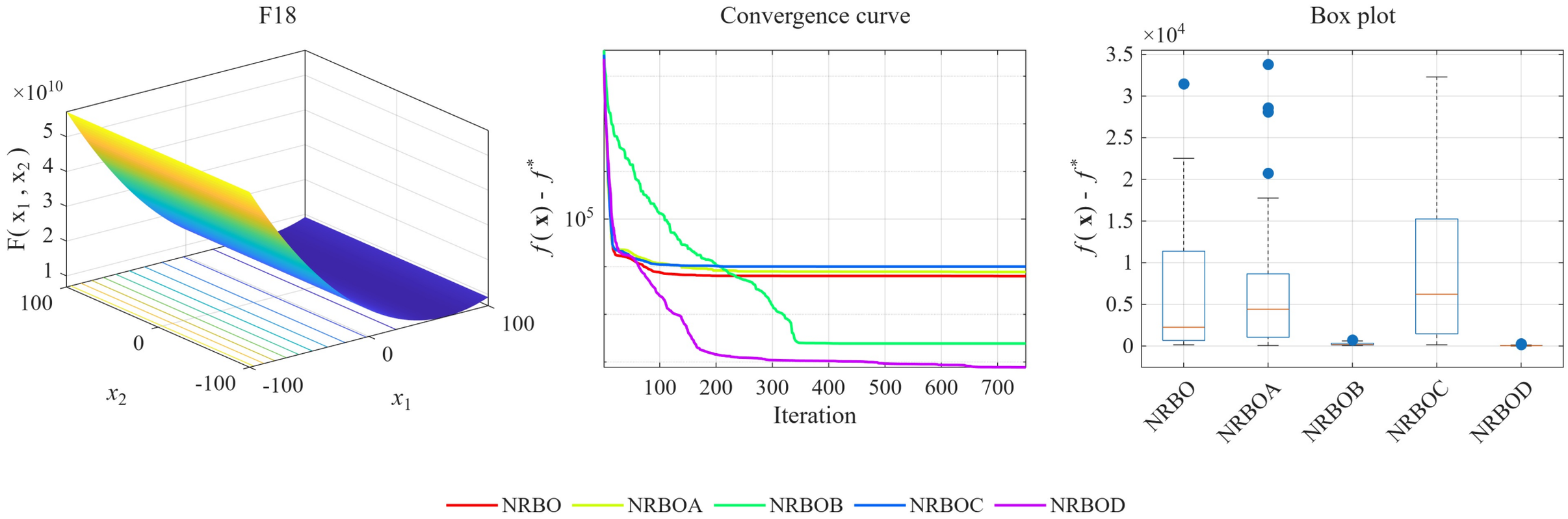

F18
×10^10
F(x1 , x2)
5
4
3
2
1
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
10^5
100
200
300
400
500
600
700
Iteration
Box plot
×10^4
f(x) - f*
3.5
3
2.5
2
1.5
1
0.5
0
NRBO
NRBOA
NRBOB
NRBOC
NRBOD
NRBO
NRBOA
NRBOB
NRBOC
NRBOD